

# Part 1 — Manual Exploratory Testing

## Defect Report

### Test Environment

|                  |                                                 |
|------------------|-------------------------------------------------|
| Application URL  | http://qainterview.pythonanywhere.com/          |
| Browser(s)       | Google Chrome (latest), Microsoft Edge (latest) |
| Operating System | Windows 11                                      |
| Testing Type     | Manual Exploratory Testing                      |
| Date of Testing  | March 2026                                      |
| Total Defects    | 9 defects identified                            |

### Defect Summary

| ID     | Title                              | Severity | Priority | Area             |
|--------|------------------------------------|----------|----------|------------------|
| DEF-01 | No Response for Negative Input     | High     | High     | Input Validation |
| DEF-02 | Infinity Result for Inputs 171–991 | High     | High     | Input Validation |
| DEF-03 | No Response for Inputs $\geq 992$  | High     | High     | Input Validation |
| DEF-04 | Inconsistent Validation Behaviour  | Medium   | High     | Input Validation |
| DEF-05 | Generic Error for Blank Input      | Medium   | Medium   | UI / Messaging   |
| DEF-06 | No Loading / Processing Feedback   | Low      | Low      | UI / UX          |
| DEF-07 | Copyright Year Duplicate           | Low      | Low      | UI / Content     |
| DEF-08 | Missing Input Guidance             | Low      | Low      | UI / UX          |
| DEF-09 | Hyperlink Navigation Observation   | Low      | Low      | Navigation       |

### Detailed Defect Reports

#### DEF-01 No Response for Negative Input

|                    |                                                                                                                                                                                                 |                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Description        | Application does not respond when a negative number is entered. No error message, visual feedback, or validation is triggered, leaving the user with no indication that their input is invalid. | SEV: High<br>PRI: High |
| Steps to Reproduce | 1. Navigate to the application URL 2. Enter -5 in the input field 3. Click Calculate 4. Observe the result area                                                                                 |                        |
| Expected Result    | Display an appropriate error message such as: "Factorial is not defined for negative numbers" or "Please enter a positive integer"                                                              |                        |
| Actual Result      | No response or feedback is displayed. The result area remains completely unchanged.                                                                                                             |                        |
| Browser(s)         | Chrome: No response. Edge: No response. Behaviour consistent across both browsers.                                                                                                              |                        |

#### DEF-02 Infinity Result for Inputs 171 to 991

|                           |                                                                                                                                                                                                              |                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Description</b>        | The application returns "Infinity" for inputs from 171 up to 991. This is caused by JavaScript IEEE 754 double-precision floating point overflow. The application does not detect or handle this gracefully. | <b>SEV: High</b><br><b>PRI: High</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Enter 171 in the input field 3. Click Calculate 4. Observe the result area 5. Repeat with values 172, 500, 991                                                         |                                      |
| <b>Expected Result</b>    | Display a controlled error message such as: "Number too large to calculate" or "Please enter a number between 0 and 170"                                                                                     |                                      |
| <b>Actual Result</b>      | Displays: "The factorial of [number] is: Infinity" for all inputs from 171 to 991.                                                                                                                           |                                      |
| <b>Browser(s)</b>         | Chrome: Infinity displayed. Edge: Infinity displayed. Behaviour consistent across both browsers.                                                                                                             |                                      |
| <b>Notes</b>              | Boundary identified: input 170 returns a valid result (7.257415615307994e+306). Input 171 is the first value to overflow.                                                                                    |                                      |

### DEF-03 No Response for Inputs >= 992

|                           |                                                                                                                                                                                                                                 |                                      |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Description</b>        | For inputs of 992 and above (including extremely large values such as 7777777777), the application produces no output and no error message. This is a distinct defect from the Infinity result — the system is entirely silent. | <b>SEV: High</b><br><b>PRI: High</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Enter 992 in the input field 3. Click Calculate 4. Observe the result area 5. Repeat with 1000, 9999, 7777777777                                                                          |                                      |
| <b>Expected Result</b>    | Display a validation message such as: "Number too large to calculate"                                                                                                                                                           |                                      |
| <b>Actual Result</b>      | No output or error message is displayed for any value >= 992. The result area remains blank.                                                                                                                                    |                                      |
| <b>Browser(s)</b>         | Chrome: No response. Edge: No response. Behaviour consistent across both browsers.                                                                                                                                              |                                      |

### DEF-04 Inconsistent Validation Behaviour Across Invalid Inputs

|                           |                                                                                                                                                                                                                                        |                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Description</b>        | Different categories of invalid input produce inconsistent responses. Blank input triggers an error, but negative numbers and out-of-range values produce no response. This inconsistency indicates incomplete input validation logic. | <b>SEV: Medium</b><br><b>PRI: High</b> |
| <b>Steps to Reproduce</b> | 1. Leave input blank and click Calculate (observe response) 2. Enter -5 and click Calculate (observe response) 3. Enter 992 and click Calculate (observe response)                                                                     |                                        |
| <b>Expected Result</b>    | All invalid inputs should trigger consistent, contextually appropriate error messages tailored to the specific input type                                                                                                              |                                        |
| <b>Actual Result</b>      | Blank: displays "Please enter an integer". Negative: no response. Value >= 992: no response. Mixed behaviour observed.                                                                                                                 |                                        |
| <b>Browser(s)</b>         | Chrome: Inconsistent. Edge: Inconsistent. Behaviour consistent across both browsers.                                                                                                                                                   |                                        |

## DEF-05 Generic Error Message for Blank Input

|                           |                                                                                                                                                                                              |                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Description</b>        | When the input field is left blank, the error message displayed is misleading. "Please enter an integer" implies the type is wrong, when the real issue is that no value was entered at all. | <b>SEV: Medium</b><br><b>PRI: Medium</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Leave the input field blank 3. Click Calculate 4. Observe the error message                                                                            |                                          |
| <b>Expected Result</b>    | Display a specific message such as: "This field is required" or "Field cannot be blank"                                                                                                      |                                          |
| <b>Actual Result</b>      | Displays: "Please enter an integer" — a generic message that does not clearly communicate that the field is empty                                                                            |                                          |
| <b>Browser(s)</b>         | Chrome: Generic error shown. Edge: Generic error shown. Behaviour consistent across both browsers.                                                                                           |                                          |

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## DEF-06 No Loading or Processing Feedback

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|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Description</b>        | The application provides no visual feedback while a calculation is being processed. There is no loading spinner, no disabled state on the button, and no indication to the user that their input is being handled. | <b>SEV: Low</b><br><b>PRI: Low</b> |
| <b>Steps to Reproduce</b> | 1. Enter any valid number (e.g. 5) 2. Click Calculate 3. Observe the button and result area during processing                                                                                                      |                                    |
| <b>Expected Result</b>    | A loading spinner is shown, or the Calculate button is temporarily disabled, to indicate processing is underway                                                                                                    |                                    |
| <b>Actual Result</b>      | No feedback is given. The button remains active and the UI is static during calculation.                                                                                                                           |                                    |
| <b>Browser(s)</b>         | Chrome: No feedback. Edge: No feedback. Behaviour consistent across both browsers.                                                                                                                                 |                                    |

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## DEF-07 Copyright Year Shows Duplicate "2026 - 2026"

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|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Description</b>        | The copyright message in the application footer displays the same year twice (e.g. "2026 - 2026"). A range format typically implies a start year and the current year. The start year appears to be incorrectly set to the current year dynamically. | <b>SEV: Low</b><br><b>PRI: Low</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Scroll to the bottom of the page 3. Observe the copyright message in the footer                                                                                                                                |                                    |
| <b>Expected Result</b>    | Copyright should display either a valid range (e.g. "2020 – 2026") or just the current year (e.g. "© 2026")                                                                                                                                          |                                    |
| <b>Actual Result</b>      | Footer displays: "2026 - 2026" — the same year repeated in a range format                                                                                                                                                                            |                                    |
| <b>Browser(s)</b>         | Chrome: Duplicate year shown. Edge: Duplicate year shown. Behaviour consistent across both browsers.                                                                                                                                                 |                                    |

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## DEF-08 Missing Persistent Input Guidance (UX)

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|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Description</b>        | The input field uses a placeholder text that disappears as soon as the user begins typing. No persistent label above the field or helper text below it is displayed, leaving users without guidance on expected input format once they start typing. | <b>SEV: Low</b><br><b>PRI: Low</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Click into the input field 3. Begin typing any value 4. Observe whether any label or guidance text remains visible                                                                                             |                                    |
| <b>Expected Result</b>    | A persistent label above the field or descriptive helper text below the field should remain visible throughout the input process                                                                                                                     |                                    |
| <b>Actual Result</b>      | The placeholder disappears on typing. No label or helper text is shown. The field appears blank with no contextual guidance.                                                                                                                         |                                    |
| <b>Browser(s)</b>         | Chrome: Placeholder disappears on type. Edge: Same behaviour.                                                                                                                                                                                        |                                    |

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## DEF-09 Hyperlink Navigation — Links Observed During Testing

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|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Description</b>        | The application contains three footer hyperlinks: "Terms and Conditions", "Privacy", and "About". These were tested during exploratory testing to verify they navigate correctly and do not result in errors or broken pages. | <b>SEV: Low</b><br><b>PRI: Low</b> |
| <b>Steps to Reproduce</b> | 1. Navigate to the application URL 2. Click the "Terms and Conditions" link and observe destination 3. Return and click "Privacy" and observe destination 4. Return and click "About" and observe destination                 |                                    |
| <b>Expected Result</b>    | All three links should navigate to their intended destination pages without errors. Pages should load successfully.                                                                                                           |                                    |
| <b>Actual Result</b>      | All three links navigated to their respective pages. No 404 errors or broken pages were observed during testing.                                                                                                              |                                    |
| <b>Browser(s)</b>         | Chrome: All links functional. Edge: All links functional.                                                                                                                                                                     |                                    |
| <b>Notes</b>              | Raised as an observation item. Links were functional at time of testing but should be included in regression test coverage to catch future breakage.                                                                          |                                    |

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